

Performance Work Statement (PWS)

Project: Turn-Key Upgrade for the Joint Base Charleston Airfield Lighting Control System (ALCS)

1.0 BACKGROUND

The 628th Civil Engineering Squadron currently operates the ADB Safegate Airfield Lighting Control System (ALCS) across two primary locations: the Main ALCS located on the Joint Base Charleston Air Base, and the North ALCS which services the North Field airfield. The core systems are running on legacy, End-of-Life Windows 10 (32-bit) operating systems and utilize unmanaged network switches. This configuration presents a cybersecurity risk and is not compliant with current Department of the Air Force security requirements. This project is a one-time technology modernization effort.

2.0 OBJECTIVE

The objective of this project is to upgrade the operating systems of the ALCS North and ALCS Main systems to a modern 64-bit architecture and replace the network hardware in order to enhance system performance, security, and long-term maintainability.

3.0 SCOPE OF WORK

The contractor shall provide all necessary personnel, materials, services, and equipment to successfully upgrade the ALCS North and ALCS Main systems as detailed in this PWS.

4.0 SPECIFIC TASKS & TECHNICAL REQUIREMENTS

4.1 TASK 1: Operating System (OS) Upgrade Service The contractor shall perform a turn-key upgrade of the ALCS Main system (Air Base) and the ALCS North system (North Field). This service must possess the following minimum characteristics:

- **4.1.1 Certified Technician:** This upgrade service must be performed by technicians who are formally certified by the OEM, ADB Safegate, to work on their proprietary ALCS.
- **4.1.2 Data Integrity:** The contractor shall perform a full backup of all existing system data, configurations, and settings prior to commencing work and a full restoration upon completion.

- **4.1.3. OS & Software Compatibility:** The contractor shall upgrade the operating system to an OEM-certified version of Windows (64-bit) that is guaranteed by ADB Safegate to be fully compatible and supportable with the proprietary ALCS software.
- **4.1.4 System Verification:** Upon completion, the contractor shall perform comprehensive system testing to verify and guarantee full operational functionality.

4.2 TASK 2: Network Infrastructure Upgrade The contractor shall supply and install new managed switches to replace the existing legacy hardware. The provided switches must meet the following minimum technical characteristics:

- **4.2.1 OEM Compatibility:** Must be certified by the OEM, ADB Safegate, as fully compatible for use within the ALCS architecture.
- **4.2.2 Security:** Must support, at a minimum, IEEE 802.1X Port-Based Network Access Control, port security (MAC-based), Access Control Lists (ACLs), and secure management via HTTPS, SSHv2, and SNMPv3. **Must provide a mechanism to tamper-proof unused physical ports in accordance with Department of the Air Force guidance.**
- **4.2.3. Hardware Specifications:** Must be a **19-inch rack-mountable (1U)** form factor providing a minimum of 12 x 10/100/1000 Mbps RJ-45 ports and 2 x 1G SFP (or 10G SFP+) uplink ports.
- **4.2.4 VLAN Support:** **Must support IEEE 802.1Q VLAN tagging.**
- **4.2.5 TAA Compliance:** Must be fully compliant with the Trade Agreements Act (TAA).

5.0 DELIVERABLES

5.1 A post-installation report detailing the work performed, system tests conducted, and confirmation of full system functionality.

5.2 All relevant technical documentation and manuals for the new hardware and software.

6.0 PERIOD OF PERFORMANCE

The work shall be completed within 90 days after contract award.

7.0 GOVERNMENT FURNISHED INFORMATION/EQUIPMENT

The Government will provide the contractor with necessary access to facilities and existing system documentation required to perform the work.